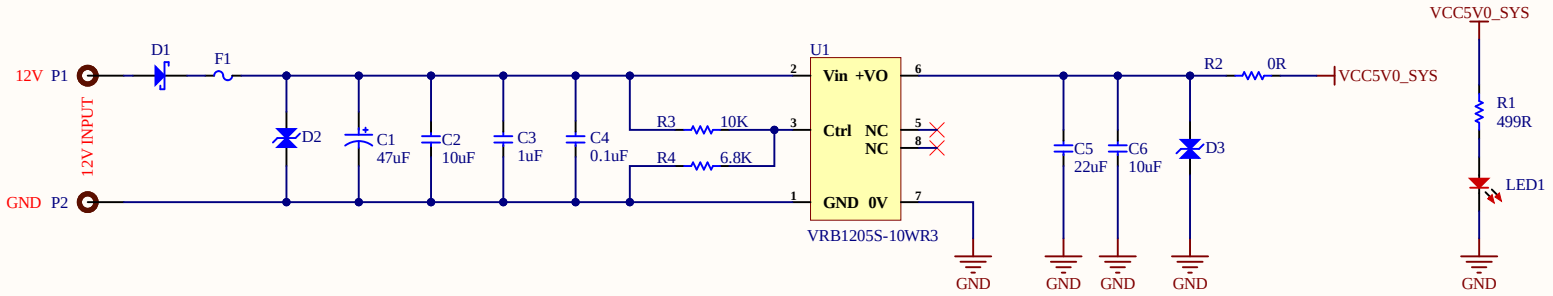
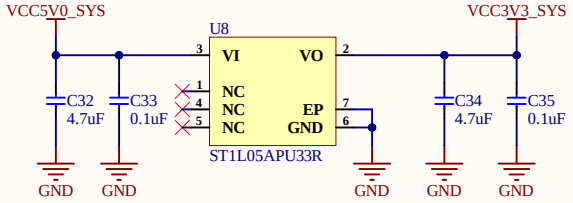
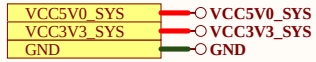
5V ISOLATED POWER MODULE

5V ISOLATED POWER MODULE



3.3V LINEAR REGULATOR

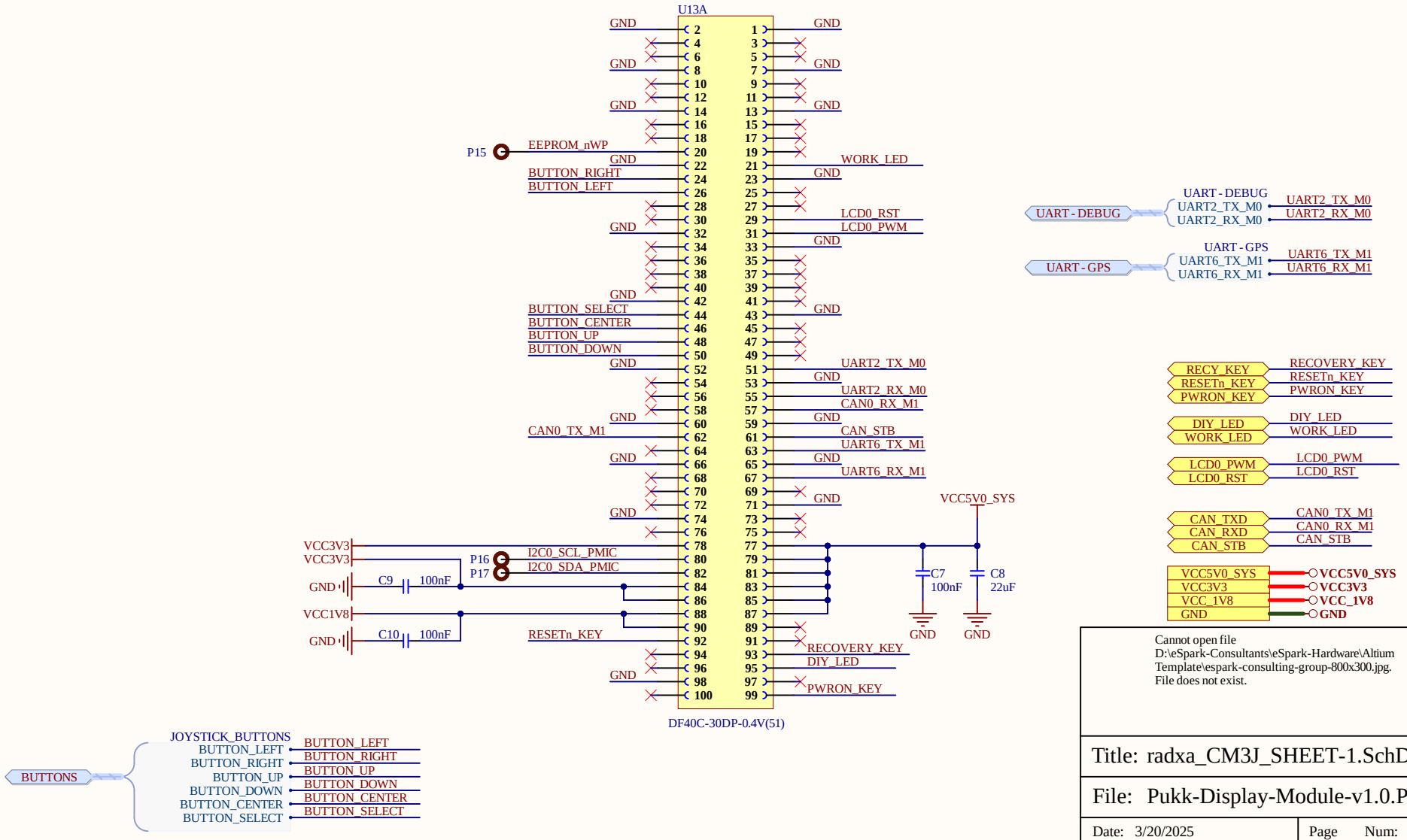




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Date: 3/24/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10

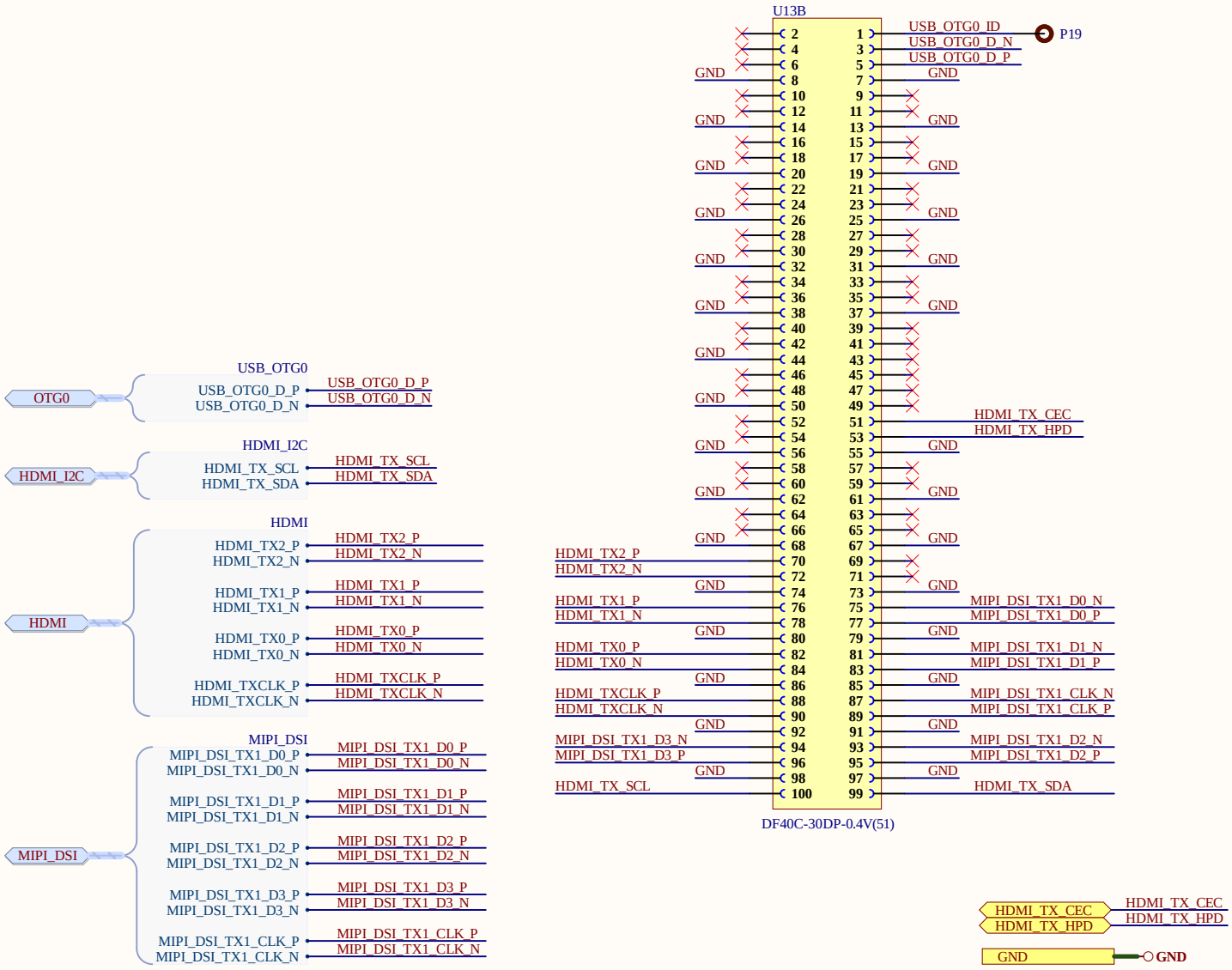
radxa_CM3J_SHEET-1



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Date: 3/20/2025	Page Num: 2
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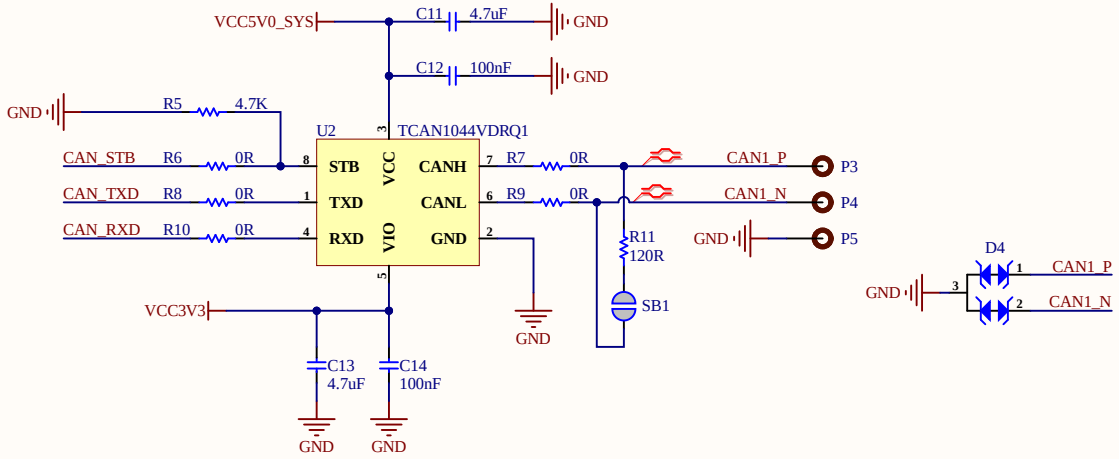
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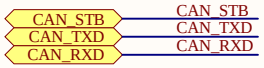


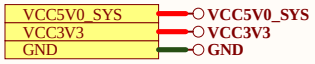
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Date: 3/19/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 10

CAN - INTERFACE - SHEET



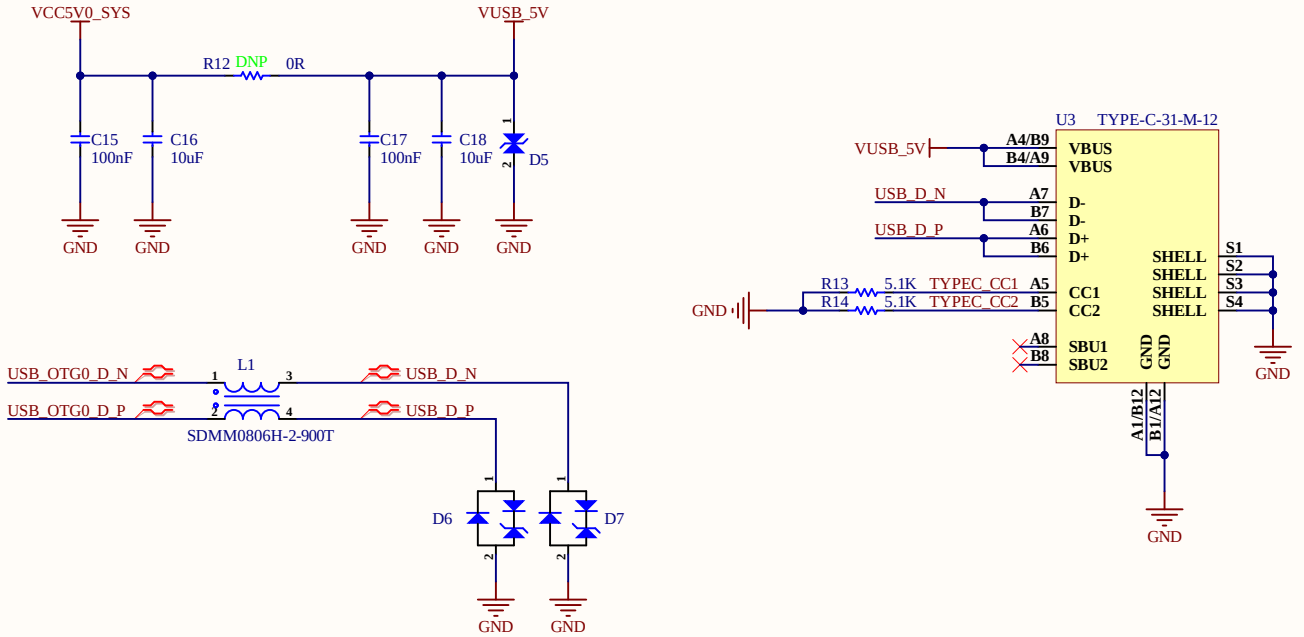


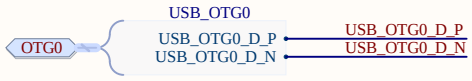


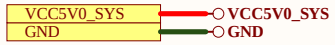
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File: Pukk-Display-Module-v1.0.PrjPcb	
Date: 3/20/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 10

TYPE-C - INTERFACE - SHEET





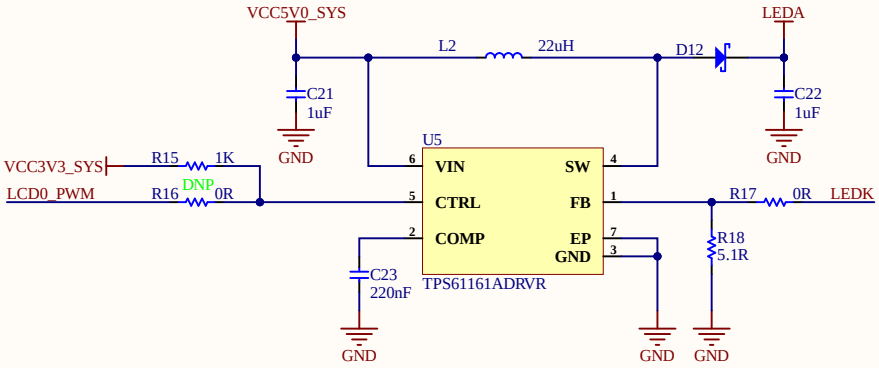


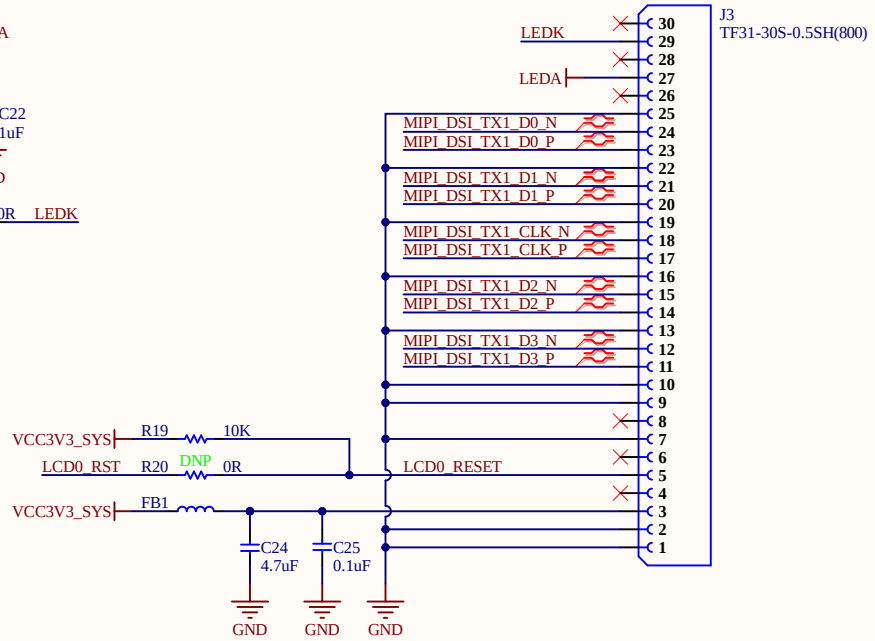
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Date: 3/20/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 10

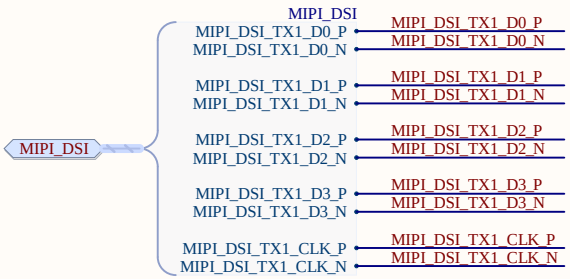
MIPI - DISPLAY - SHEET

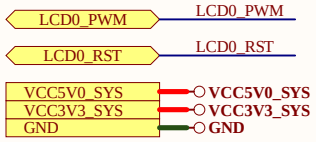
KD050FHFID103





Pin Name
NC
LEDK
NC
LEDA
NC
GND
MIPI_D0N
MIPI_D0P
GND
MIPI_D1N
MIPI_D1P
GND
MIPI_CLN
MIPI_CLP
GND
MIPI_D2N
MIPI_D2P
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NC
RESET
NC
VCI
GND
GND

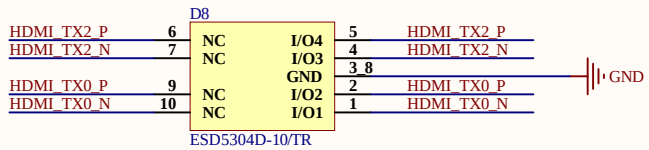


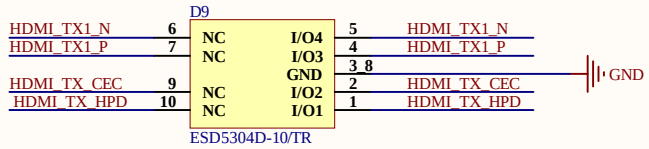


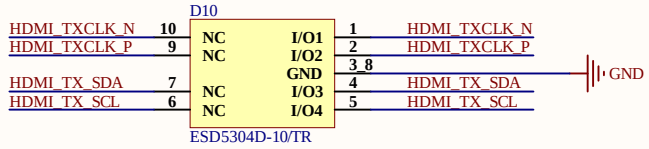
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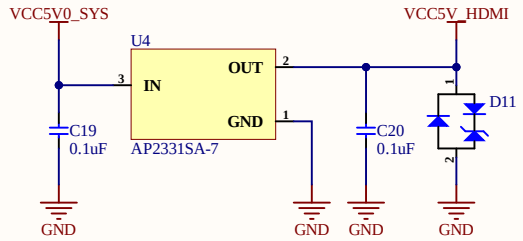
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Date: 3/20/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10

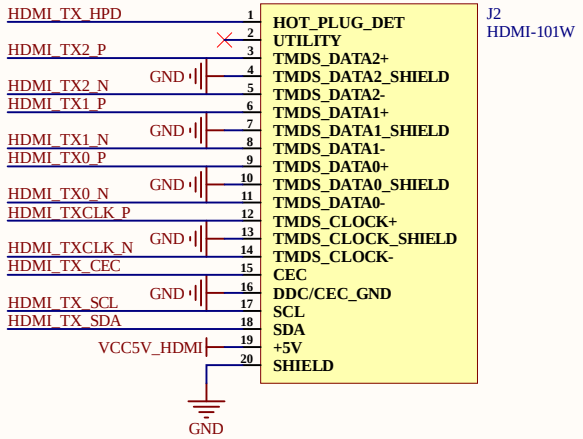
HDMI2.0 - INTERFACE - SHEET

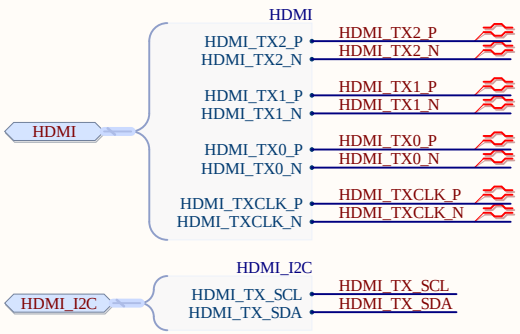


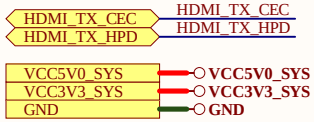








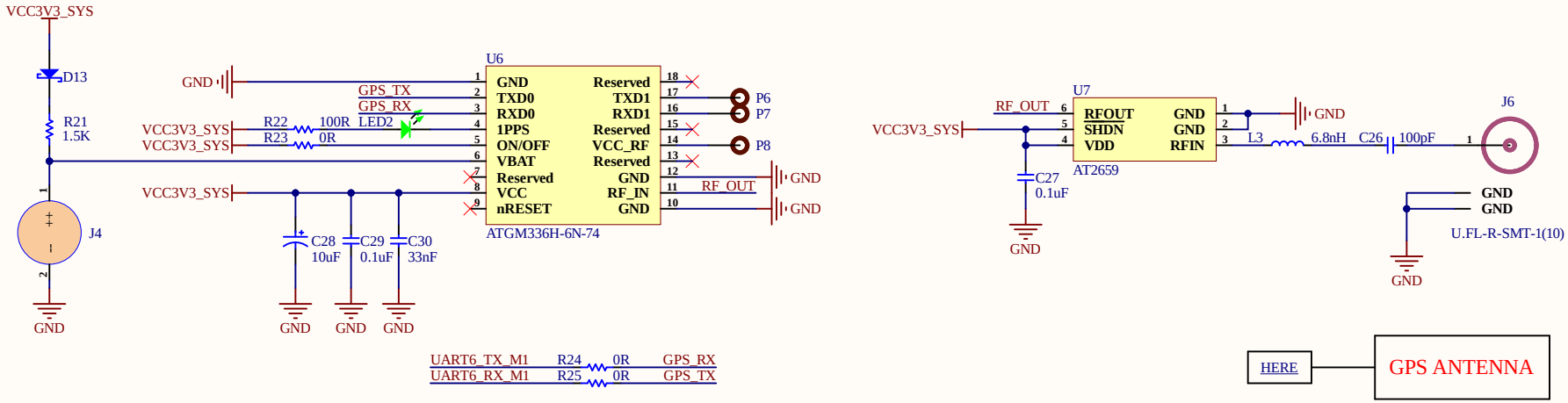


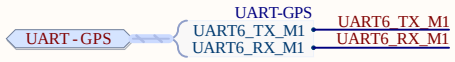


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Title: HDMI2.0_SHEET.SchDoc	
File: Pukk-Display-Module-v1.0.PrjPcb	
Date: 4/7/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 10

GPS - INTERFACE - SHEET



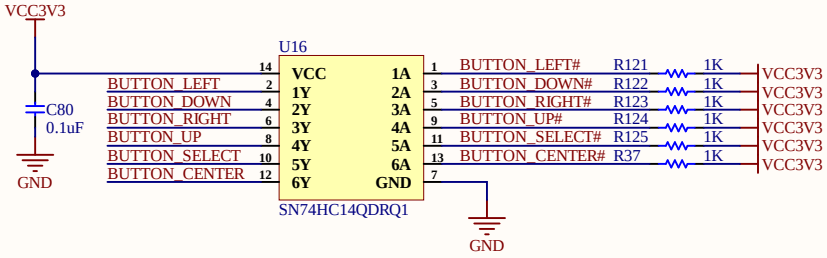


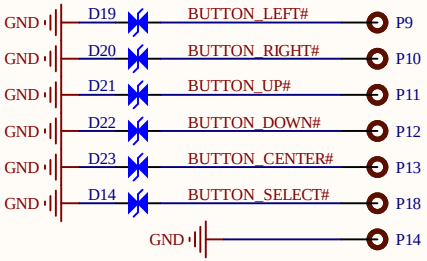


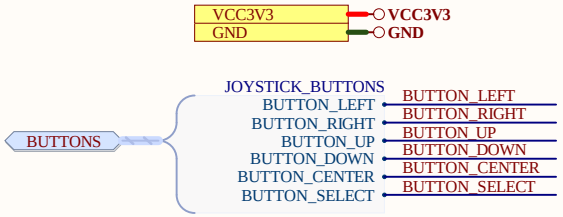
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Title: GPS_INTERFACE_SHEET.SchDoc	
File: Pukk-Display-Module-v1.0.PrjPcb	
Date: 3/19/2025	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 10

JOYSTICK - BUTTON - SHEET





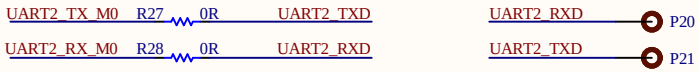


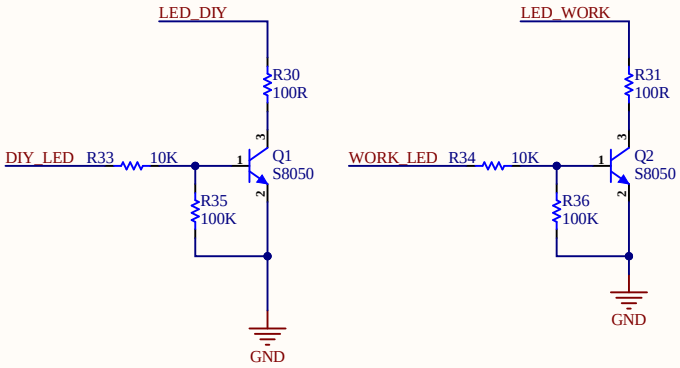
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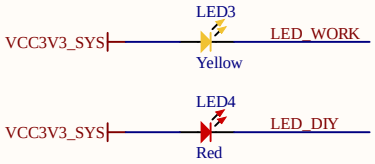
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File: Pukk-Display-Module-v1.0.PrjPcb	
Date: 4/7/2025	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 10

MISCELLANEOUS - SHEET

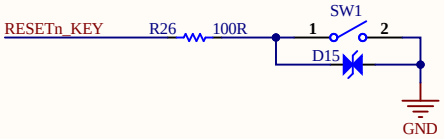
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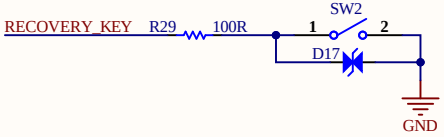




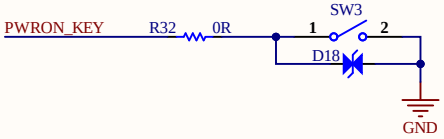
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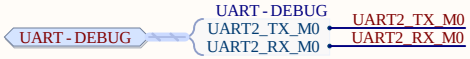


BOOT



POWER



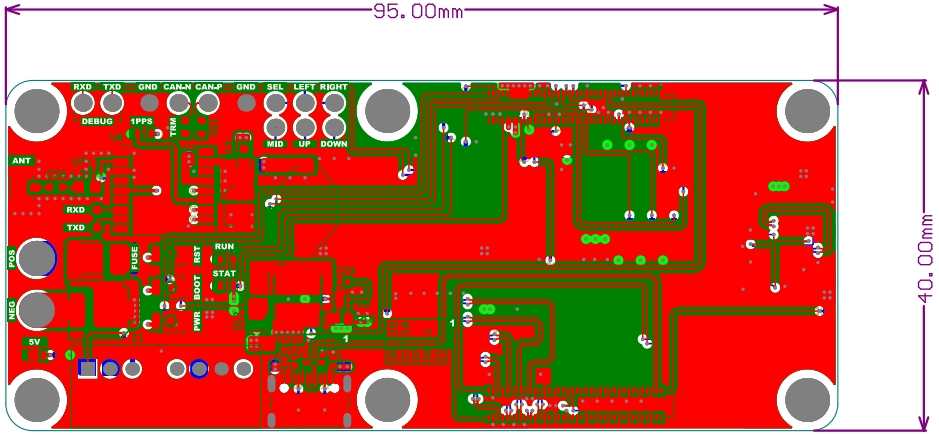






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Title: MISCELLANEOUS_SHEET.SchDoc	
File: Pukk-Display-Module-v1.0.PrjPcb	
Date: 3/25/2025	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 10



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	4	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.203mm	4.74	
2	GND	CF-004	0.030mm		
	Dielectric 1	FR-4	1.030mm	4.6	
3	POWER	CF-004	0.030mm		
	Dielectric 3	PP-006	0.203mm	4.74	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	4	
	Bottom Overlay				
Total board thickness:			1.586mm		

Comment	Description	Designator	Footprint	LibRef	Quantity
47uF	CAP ALUM POLY 47UF 20% 25V SMD	C1	25SVPF47M	25SVPF47M	1
10uF	CAP CER 10UF 25V X5R 0603	C2, C6	Cap-0603	GRM188R61E106MA73D	2
1uF	CAP CER 1UF 25V X7R 0603, CAP CER 1UF 50V X5R 0603	C3, C21, C22	Cap-0603	C1608X7R1E105K080AB, CL10A105KB8NNNC	3
0.1uF	CAP CER 0.1UF 25V X7R 0603	C4	Cap-0603	CC0603JRX7R8BB104	1
22uF	CAP CER 22UF 10V X7R 0805	C5	Cap-0805	GRM21BZ71A226ME15L	1
100nF	CAP CER 0.1UF 16V X5R 0402	C7, C9, C10, C12, C14, C15, C17	Cap-0402	CC0402KRX5R7BB104	7
22uF	CAP CER 22UF 10V X5R 0402	C8	Cap-0402	GRM158R61A226ME15D	1
4.7uF	CAP CER 4.7UF 10V X5R 0402, CAP CER 4.7UF 16V X5R 0402	C11, C13, C24, C32, C34	Cap-0402	CC0402KRX5R6BB475, CL05A475MO5NUNC	5
10uF	CAP CER 10UF 10V X5R 0402	C16, C18	Cap-0402	CC0402MRX5R6BB106	2
0.1uF	CAP CER 0.1UF 50V X7R 0402, CAP CER 0.1UF 10V X5R 0402	C19, C20, C25, C27, C29, C33, C35, C80	Cap-0402	GRM155R71H104ME14D, CC0402KRX5R6BB104	8
220nF	CAP CER 0.22UF 25V X5R 0402	C23	Cap-0402	CC0402KRX5R8BB224	1
100pF	CAP CER 100PF 50V C0G/NPO 0603	C26	Cap-0603	CC0603JRNPO9BN101	1
10uF	CAP ALUM 10uF 25V SMD	C28	RVT1E100M0405_	RVT1E100M0405	1
33nF	CAP CER 0.033UF 25V X7R 0402	C30	Cap-0402	CC0402KRX7R8BB333	1
SS56	DIODE SCHOTTKY 60V 5A SMA	D1	SMA	SS56	1
SD12C.TCT	24V 350W 13.3V 12V SOD-323 ESD and Surge Protection (TVS/ESD) ROHS	D2	SOD-323 (Bi-directional)	SD12C.TCT	1
SMAJ5.0CA	TVS DIODE 5VWM 9.2VC SMA	D3	SMA Bi Directional	SMAJ5.0CA	1
CDSOT23-T24CAN-Q	TVS DIODE 24VWM 40VC SOT23-3	D4	SOT-23	CDSOT23-T24CAN-Q	1
ESD5451N	DFN1006-2(0402) ESD Protection Devices ROHS	D5, D14, D15, D17, D18, D19, D20, D21, D22, D23	DFN1006-2L (Bi-Directional)	ESD5451N	10
ESD5341N-2/TR	4A@8/20us 15V 7V 5V Unidirectional DFN1006-2L TVS ROHS	D6, D7, D11	DFN1006-2L_	ESD5341N-2/TR	3
ESD5304D-10/TR	4A@8/20us 12V 7V 5V Unidirectional DFN2510-10L TVS ROHS	D8, D9, D10	DFN2510-10L_	ESD5304D-10/TR	3
SS16	60V 700mV@1A 1A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D12	SMA	SS16	1
BSB19WS	DIODE SCHOTTKY 40V 1A SOD323	D13	SOD-323	BSB19WS	1
BSMD1812-150-33V	33V 1.5A 40A 3A 1812 Resettable Fuses ROHS	F1	Fuse-1812_	BSMD1812-150-33V	1
BLM18PG181SN1D	0603 180Ω ±25% DCR=0.09Ω 1.5A	FB1	IND-0603 (1608 Metric)	BLM18PG181SN1D	1
HDMI-101W	SMD HDMI Connectors ROHS	J2	HDMI-101W	HDMI-101W	1
TF31-30S-0.5SH(800)	CONN FFC BOTTOM 30POS 0.5MM R/A	J3	TF31-30S-0.5SH(800)	TF31-30S-0.5SH(800)	1
PB-621SR13-TF	SMD Button And Strip Battery Connector ROHS	J4	PB-621SR13-TF	PB-621SR13-TF	1
IJ.FL-R-SMT-1(10)	CONN U.FL RCPT STR 50 OHM SMD	J6	IJ.FL-R-SMT-1(10)	IJ.FL-R-SMT-1(10)	1
SDMM0806H-2-900T	900Ω@100MHz 4Ω 2 0302 Common Mode Filters ROHS	L1	0302	SDMM0806H-2-900T	1
22uH	FIXED IND 22UH 750MA 496MOHM SMD	L2	VLCF5020T-220MR75-1	VLCF5020T-220MR75-1	1
6.8nH	600mA 6.8nH ±5% 110mΩ 0805 Inductors (SMD) ROHS	L3	Ind-0805	BKWO805U06N8JGT	1
RED	RED LED 0603 Light Emitting Diodes	LED1, LED4	LED-0603	NCD0603R3	2
Emerald	Emerald 0603 Light Emitting Diodes (LED) ROHS	LED2	LED-0603	NCD0603G2	1
Yellow	yellow 0603 Light Emitting Diodes (LED) ROHS	LED3	LED-0603	NCD0603Y2	1
S8050	Trans Gp Bjt NPN 25V 1.5A 300MW 3-PIN SOT-23 T/r	Q1, Q2	SOT-23	S8050	2
499R	RES 499 OHM 1% 1/16W 0402	R1	Res-0402	RC0402FR-07499RL	1
0R	RES 0 OHM JUMPER 1/8W 0805	R2	Res-0805	RC0805FR-070RL	1
10K	RES 10K OHM 1% 1/16W 0402	R3, R19, R33, R34	Res-0402	RC0402FR-0710KL	4
6.8K	RES SMD 6.8K OHM 0.1% 1/16W 0402	R4	Res-0402	RT0402BRC076K8L	1
4.7K	RES 4.7K OHM 1% 1/16W 0402	R5	Res-0402	RC0402FR-074K7L	1
0R	RES 0 OHM JUMPER 1/16W 0402	R6, R7, R8, R9, R10, R17, R23, R24, R25, R27, R28, R32	Res-0402	RC0402FR-070RL	12
120R	RES 120 OHM 1% 1/10W 0603	R11	Res-0603	RC0603FR-07120RL	1
DNP	RES 0 OHM JUMPER 1/8W 0805	R12	Res-0805	RC0805FR-070RL	1
5.1K	RES 5.1K OHM 1% 1/16W 0402	R13, R14	Res-0402	RC0402FR-075K1L	2
1K	RES 1K OHM 1% 1/16W 0402	R15, R37, R121, R122, R123, R124, R125	Res-0402	RC0402FR-071KL	7
DNP	RES 0 OHM JUMPER 1/16W 0402	R16, R20	Res-0402	RC0402FR-070RL	2
5.1R	RES 5.1 OHM 5% 1/8W 0805	R18	RES-0805	RC0805JR-075R1L	1
1.5K	RES 1.5K OHM 1% 1/16W 0402	R21	Res-0402	RC0402FR-071K5L	1
100R	RES 100 OHM 1% 1/16W 0402	R22, R26, R29, R30, R31	Res-0402	RC0402FR-07100RL	5
100K	RES 100K OHM 1% 1/16W 0402	R35, R36	Res-0402	RC0402FR-07100KL	2
B3U-1000P	SWITCH TACTILE SPST-NO 0.05A 12V	SW1, SW2, SW3	B3U-1000P	B3U-1000P	3
VRB1205S-10WR3	SIP Isolated Power Modules ROHS	U1	VRB1205S-10WR3	VRB1205S-10WR3	1
TCAN1044VDRQ1	IC TRANSCEIVER 1/1 8SOIC	U2	SOIC-8(0.154", 3.90mm Width)	TCAN1044VDRQ1	1
TYPE-C-31-M-12	5A 1 16P Female Type C SMD USB Connectors ROHS	U3	TYPE-C-31-M-12_	TYPE-C-31-M-12	1
AP2331SA-7	IC CURRENT SWITCH SOT23-3	U4	SOT-23	AP2331SA-7	1
TPS61161ADRVR	IC LED DRV RGLTR PWM 700MA 6WSON	U5	WSON-6-EP(2x2)	TPS61161ADRVR	1
ATGM336H-6N-74	SMD, 10.1x9.7mm Satellite Positioning Modules ROHS	U6	ATGM336H-6N-74	ATGM336H-6N-74	1
AT2659	1.4V~3.6V 4.3mA DFN-6(1x1.6) Low Noise Amplifiers (LNA) - RF ROHS	U7	DFN-6(1x1.6)	AT2659	1
ST1105APU33R	IC REG LINEAR 3.3V 1.3A 6DFN	U8	DFN-6(3x3)	ST1105APU33R	1
DF40C-30DP-0.4V(51)	CONN RCPT 100POS SMD GOLD	U13A, U13B	DF40C-100DS-0.4V(51)	DF40C-30DP-0.4V(51)	2
SN74HC14QDRQ1	IC INVERT SCHMITT 6CH 1/N 14SOIC	U16	SOIC-14	SN74HC14QDRQ1	1